WORKED SOLUTION

STEP I 2016, Question 9

Mechanics • Static equilibrium: wall and rod

STEP I 2016, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q9

The key idea

Take moments about A: the two forces at the wall then drop out and the normal reaction at the rail comes straight out in terms of θ, with no friction involved. Resolving horizontally and vertically finishes the job, once we know which way each friction force acts. That is the real content of the question: friction opposes the motion the rod is about to make, so we must decide whether the end A is about to slide down the wall or up it.

Solution

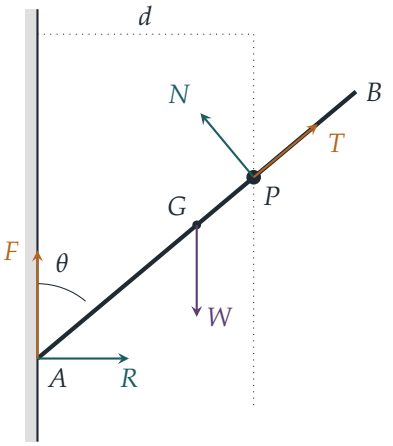
Model the rod as a uniform rigid rod of weight W, with centre G at distance a from A, and the rail as a fixed line meeting the rod at a single point P. Since the horizontal distance from A to the rail is d and the rod makes angle θ with the vertical,

$$AP = \frac{d}{\sin\theta}.$$

The condition $a\sin\theta < d$ says that G lies between A and P.

The forces are:

- at A: a normal reaction R, horizontal and away from the wall, and friction F along the wall;
- at P: a normal reaction N perpendicular to the rod, and friction T along the rod;
- the weight W at G.



The arrows for F and T show the directions for the first case below.

Moments about A

The moment of N about A is $N \cdot AP$, since N is perpendicular to the rod; T acts along the rod and has no moment about A, and neither do R and F. The weight has lever arm $a\sin\theta$. So

$$N\frac{d}{\sin\theta} = Wa\sin\theta \quad\Longrightarrow\quad N = \frac{Wa\sin^2\theta}{d}. \tag{1}$$

This holds whichever way the rod is about to slip.

Which way does it slip?

> **Ask yourself** Is A about to slide down the wall or up it?

Suppose first that there were no friction at all. Resolving vertically would need $N\sin\theta = W$, and with (1) this means $a\sin^3\theta = d$. Here $a\sin^3\theta < a\sin\theta < d$, so by (1) $N\sin\theta < W$: the normal reactions alone cannot hold the rod up, and friction must supply the rest. The rod is about to slip *downwards*: A slides down the wall, θ decreases, $AP = d/\sin\theta$ grows, and so the rod slides down through the rail towards the wall. Friction opposes this, so F acts up the wall and T acts up the rod (towards B), as drawn. In limiting equilibrium at both contacts,

$$F = \mu R, \qquad T = \lambda N.$$

Resolving

The unit vector up the rod is $(\sin\theta, \cos\theta)$ and the direction of N is $(-\cos\theta, \sin\theta)$ (towards the wall and upwards). Resolving horizontally:

$$R - N\cos\theta + \lambda N\sin\theta = 0 \quad\Longrightarrow\quad R = N(\cos\theta - \lambda\sin\theta).$$

(For the wall to push, we need $R \geq 0$, so this case requires $\tan\theta \leq 1/\lambda$.) Resolving vertically:

$$\begin{aligned} W &= \mu R + N\sin\theta + \lambda N\cos\theta \\ &= N\big(\mu\cos\theta - \lambda\mu\sin\theta + \sin\theta + \lambda\cos\theta\big) \\ &= N\big((\lambda+\mu)\cos\theta + (1-\lambda\mu)\sin\theta\big). \end{aligned}$$

Substituting N from (1) and dividing by $W\sin^2\theta$,

$$d\,\mathrm{cosec}^2\theta = a\big((\lambda+\mu)\cos\theta + (1-\lambda\mu)\sin\theta\big),$$

which is the required result.

The case $a\sin\theta > d$

Now G lies beyond the rail. Equation (1) is unchanged, as is the geometry, so the only question is again the direction of slipping.

Ask yourself What changes now that G is beyond the rail?

The argument above showed that downward slipping happens when $a\sin^3\theta < d$, and upward slipping when $a\sin^3\theta > d$. With $a\sin\theta < d$ only the first was possible. With $a\sin\theta > d$ the second becomes possible: the overhanging weight beyond the rail makes $N\sin\theta > W$, so the rod tends to slide down through the rail *away* from the wall, with A moving up the wall. This is the new case, and in it both friction forces are reversed: $F = \mu R$ acts down the wall and $T = \lambda N$ acts down the rod. Replacing λ by $-\lambda$ and μ by $-\mu$ in the working above,

$$R = N(\cos\theta + \lambda\sin\theta), \qquad W = N\big((1-\lambda\mu)\sin\theta - (\lambda+\mu)\cos\theta\big),$$

and with (1),

$$d\,\mathrm{cosec}^2\theta = a\big((1-\lambda\mu)\sin\theta - (\lambda+\mu)\cos\theta\big).$$

Here $R > 0$ automatically. (If instead $a\sin^3\theta < d < a\sin\theta$, the rod is still on the point of slipping downwards and the first result applies unchanged, since its derivation never used where G was.)

Checks

- With no friction ($\lambda = \mu = 0$) both results reduce to $d = a\sin^3\theta$, the frictionless balance found above.
- Writing $\lambda = \tan\alpha$ and $\mu = \tan\beta$ (the angles of friction), the right-hand sides become $a\sin(\theta+\alpha+\beta)/(\cos\alpha\cos\beta)$ and $a\sin(\theta-\alpha-\beta)/(\cos\alpha\cos\beta)$: the two cases differ only in the sign of the friction angles, as they should.
- With $\lambda = \mu = \frac{1}{2}$ and $\theta = \frac{1}{3}\pi$, the two cases give $d/a = \frac{3}{8} + \frac{9\sqrt{3}}{32} \approx 0.862$ and $d/a = \frac{9\sqrt{3}}{32} - \frac{3}{8} \approx 0.112$. Substituting each back into the force and moment equations, with the friction directions above, gives zero resultant and zero moment.

Common slips

- Taking the rail's normal reaction to be vertical. It is perpendicular to the rod, because the rod touches the rail at a single point.
- Using the length d instead of $d / \sin\theta$ as the distance from A to the rail in the moment equation.
- Fixing the friction directions without asking which way the rod would actually move.

Beyond STEP

The rod is stable for a whole interval of angles θ, between the downward and upward limiting cases: friction gives a range of equilibria, not a single one. Describing this kind of set (the friction cone at each contact, and the set of states the contacts can hold) is the starting point of contact mechanics and of the analysis of grasping in robotics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body statics carefully at first-year level.

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